

Password Security Assessment & Risk Analysis Tool

1. Introduction

This project is a Password Security Analyzer developed in Assembly Language using Irvine32 library. It evaluates user entered passwords based on security rules such as character variety, length, and complexity. The system classifies passwords into Weak, Medium, or Strong categories and also provides security feedback.

2. Objective

The main objective of this project is to:

- Analyze password strength using basic security rules
- Identify missing security elements (uppercase, digits, symbols)
- Provide a risk-based classification of passwords
- Introduce the concept of brute-force vulnerability in passwords

3. Tools & Technologies Used

- Assembly Language (x86)
- Irvine32 Library
- Visual Studio

4. Working of the System

The system works in the following steps:

1. User enters a password
2. Program scans each character
3. It checks for:
 - Uppercase letters
 - Lowercase letters
 - Digits
 - Special characters
4. It assigns a score based on detected features

5. Final strength is calculated: weak, medium and hard.
6. It also displays reasons and brute-force risk level

5. Test Cases & Outputs

Test Case 1: Weak Password

Input Example: 12345

Result:

- Missing uppercase
- Missing special character
- Too short password
- Brute Force Risk: Can be cracked quickly

```
Enter Password: 12345

Missing uppercase
Missing lowercase
Missing special character
Too short password

Result: WEAK
Brute Force Risk: Can be cracked quickly
PS C:\Users\DELL 8\OneDrive\Desktop\coal> |
```

Test Case 2: Medium Password

Input Example: AS234DFXC45

Result:

- Missing special character
- Moderate strength
- Brute Force Risk: May take time to crack

```
Enter Password: AS234DFxc45

Missing special character

Result: MEDIUM
Brute Force Risk: May take time to crack _
```

Test Case 3: Strong Password

Input Example: Qw9@Lm7#Zp

Result:

- All conditions satisfied
- Strong password
- Brute Force Risk: Very hard to crack

```
Enter Password: Qw9@Lm7#Zp

Result: STRONG
Brute Force Risk: Very hard to crack
PS C:\Users\DELL 8\OneDrive\Desktop\coal> █
```

6. Conclusion

This project successfully demonstrates password strength analysis using Assembly Language. It improves understanding of character classification, conditional logic, and basic cybersecurity principles such as password complexity and bruteforce resistance. The system helps users understand why a password is weak or strong, rather than only labeling it.